Vertically Opposite Angles

Year 6 · Mathematics · Measurement | C4 · Learn + Notice

Name: _____ Date: _____

LEARNING INTENTION

I can distinguish opposite and adjacent angles at an intersection.

KEY IDEA

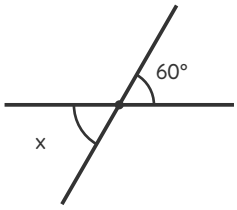
When two straight lines cross, vertically opposite angles are equal. Adjacent angles on a straight line add to 180°.

VOCABULARY

intersection · vertically opposite · adjacent · supplementary

LEARN IT: A Worked Model

One angle at an intersection is 60°. The opposite angle is also 60°. Each adjacent angle is 180° - 60° = 120°.



Use the marked opening. Diagrams for calculation tasks are not drawn to scale.

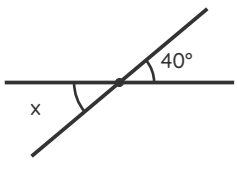
SUCCESS CRITERIA

- ☐ I can distinguish opposite and adjacent angles at an intersection.
- ☐ I can explain the benchmark, tool or angle rule I used.
- ☐ I can check my answer and correct a mistake.

NOTICE IT: Guided Practice

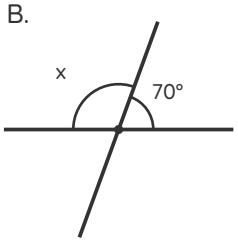
Find x. Write the rule or total you used.

A.



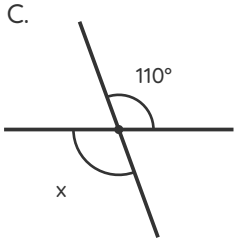
Answer: _____

B.



Answer: _____

C.



Answer: _____

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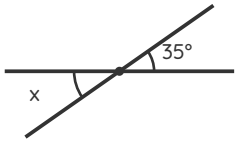
Year 6 · Mathematics · Measurement | C4 · Practise + Apply + Reason

PRACTISE IT: Independent Questions

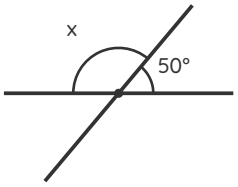
Find x. Write the rule or total you used.

A.



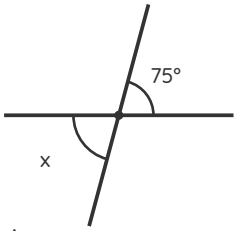
Answer: _____

B.



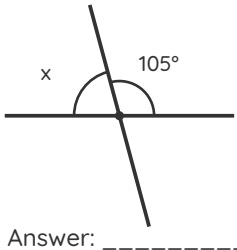
Answer: _____

C.



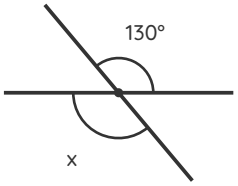
Answer: _____

D.



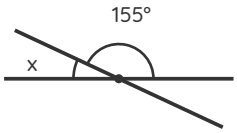
Answer: _____

E.



Answer: _____

F.



Answer: _____

APPLY IT

Two straight paths cross. One angle is 68°. Find the other three angles in order around the intersection.

EXPLAIN HOW YOU KNOW: Misconception Check

Zoe says all four angles at an intersection are equal. When would that actually be true?

CHALLENGE: Reasoning Task

An angle equals its adjacent angle at an intersection. Find both angles and explain.

MY SELF-REVIEW

Circle one: I understand it / I need more practice / I need help

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ANSWER KEY: Guided + Independent Practice

Notice: A 40°; B 110°; C 110°.

Practice: A 35°; B 130°; C 75°; D 75°; E 130°; F 25°.

Apply: adjacent 112°, opposite 68°, adjacent 112°.

Reason: all four are equal only when the lines are perpendicular, making four 90° angles.

Challenge: both are 90°, since equal adjacent angles total 180°.

COMMON MISCONCEPTION

Vertically opposite describes a position across the vertex, not two angles that happen to face upwards/downwards.

TEACHER PROMPT

Ask: "Which opening are you using? What benchmark or angle relationship applies? How can you check your answer?"

NEXT-STEP GUIDANCE

Move to mixed angle reasoning.

PRINTING AND ASSESSMENT

Print A4. Diagrams are not to scale for calculation purposes. Use stated values and angle relationships; check the explanation as well as the result.

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